Techniques to impove battery energy storage profits on the Belgian Imbalance Market

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December 2025

1 Problem Formulation

Businesses often purchase their expected electricity needs on the day-ahead market. However, their actual consumption on the day of delivery can differ from what was planned. This difference, or *imbalance*, must be settled on the imbalance market. The prices in the imbalance market are significantly higher compared to the day-ahead market. Furthermore, the price is only decided at the end of a quarter, so the price for the electricity used in the past quarter will only be calculated based on the amount used during the entire quarter, multiplied by the price of the last minute of the quarter . Every minute, a forecast of the last quarter can be obtained, but this is not necessarily a good one.

To mitigate the financial risks associated with this price volatility, businesses can use a Battery Energy Storage System (BESS). A battery allows a business to store energy when prices are low (or even negative) and use that stored energy when prices are high. This avoids the need to buy expensive electricity from the grid to cover an imbalance.

The challenge is to develop an intelligent control policy for such a battery that minimizes the cost of electricity. We will discover a range of different families of techniques that can be tried for this problem.

2 Assumptions

3 Baseline Heuristic

To establish a minimum performance benchmark, a simple heuristic policy was developed. This policy relies on intuitive, transparent rules, providing a benchmark for evaluating the more complex strategies introduced later in this work.

3.1 The Model

Furthermore, this section serves to introduce the formal notation and state-space formulation that will be used consistently throughout this document.

3.1.1 Variables

In this section, the variables used in the model components are presented and clarified.

SoC_t The amount of energy (in MWh) stored in the battery at time t.

p_t The imbalance price (€/MWh) on the electricity grid at time t.

x_t The amount of power (in MW) bought from the grid ($x_t > 0$) or sold to the grid ($x_t < 0$) during the interval at time t.

A_t^c, A_t^d Accumulated energy (in MWh) traded within a 15-minute quarter. They are reset to zero at the beginning of a new quarter.

r^{chg}, r^{dis} maximum charge/discharge rate in (MW).

3.1.2 Model Components

1. **State Variable (S_t)**
   The state at time t consists of the current battery state of charge, the price of electricity at time t, which is a forecast if $(t+1) \mod 15 \neq 0$:

$$S_t = (SoC_t, p_t, A_t^c, A_t^d,)$$

2. **Decision Variable (x_t)**
   The decision variable x_t represents the net energy flow to or from the battery during the time interval at time t. A positive value ($x_t > 0$) indicates charging, while a negative value ($x_t < 0$) indicates discharging. This decision is subject to the physical constraints of the battery system.

   Let R_t be the energy stored in the battery at the beginning of the interval, R^{max} be the maximum battery capacity, and r^{chg} and r^{dis} be the maximum charge and discharge rates (in energy per time interval), respectively. The decision x_t must satisfy the following constraints:

$$\begin{aligned}
x_t &\leq SoC^{max} - SoC_t && \text{(Cannot charge more than available capacity)} \\
x_t &\geq -SoC_t && \text{(Cannot discharge more than stored energy)} \\
x_t &\leq r^{chg} && \text{(Cannot charge faster than the maximum charge rate)} \\
x_t &\geq -r^{dis} && \text{(Cannot discharge faster than the maximum discharge rate)}
\end{aligned}$$

3. **Exogenous Information (W_{t+1})**
   This is the new, unknown information that arrives after decision x_t is made.
   $$W_{t+1} = (p_{t+1})$$

4. **Transition Function**
   $$SoC_{t+1} = SoC_t + x_t$$
   $$A^c_{t+1} = \begin{cases} \max(0, x_t) & \text{if } (t+1) \pmod{15} = 0 \\ A^c_t + \max(0, x_t) & \text{otherwise} \end{cases}$$
   $$A^d_{t+1} = \begin{cases} \max(0, -x_t) & \text{if } (t+1) \pmod{15} = 0 \\ A^d_t + \max(0, -x_t) & \text{otherwise} \end{cases}$$
   The price p_{t+1} is observed after the transition and is independent of the action x_t. This gives the new state vector:
   $$S_{t+1} = (SoC_{t+1}, p_{t+1}, A^c_{t+1}, A^d_{t+1})$$

5. **Reward Function ($R(S_t, x_t)$)**
   This function defines the immediate financial outcome of an action taken at time t. Due to market rules, financial settlement occurs only at the end of each 15-minute interval. This results in a sparse reward signal: the intermediate contribution is zero, and the accumulated profit or loss is realized in the final minute of the quarter.

   We define the reward as the financial profit (revenue minus cost):
   $$R(S_t, x_t) = \begin{cases} -p_t \cdot (A^c_{t+1} - A^d_{t+1}) & \text{if } (t+1) \pmod{15} = 0 \\ 0 & \text{otherwise} \end{cases}$$
   where a positive R indicates profit (money earned) and a negative R indicates a loss.

6. **Objective Function**
   The goal of the agent is to find an optimal policy π^* that maximizes the expected cumulative discounted reward over an episode of length T. Formally, the objective function $J(\pi)$ is defined as:
   $$\max_\pi J(\pi) = \max_\pi \mathbb{E}\left[\sum_{t=0}^{T} \gamma^t R(S_t, x_t)\right]$$
   where:
   - $\mathbb{E}$ denotes the expected value given the stochastic nature of the environment (prices).
   - $\gamma \in [0, 1]$ is the discount factor, determining the importance of future rewards.

3.2 State Space Decisions

In this section some extra context is given to the decisions of adding certain features.

3.2.1 A_t^c, A_t^d: Tracking Intra-Quarter Trading

The original state representation, $S_t = (SoC_t, p_t)$, was found to be insufficient for the agent to properly assign credit for its actions within a 15-minute settlement period. This created a critical ambiguity, which can be illustrated by comparing two scenarios:

- **Case 1:** The agent starts a 15-minute period with a full battery. It remains idle for 14 minutes and then discharges in the final minute. The reward for this quarter is positive, calculated as $R \approx x_{discharge} \cdot p_{15}$.
- **Case 2:** The agent starts the period with an empty battery. It charges the battery full in the first minute, holds the charge, and then discharges in the final minute. The reward for this quarter is significantly lower (likely negative), as it is based on the net energy traded: $R \approx (-C_{capacity} + x_{discharge}) \cdot p_{15}$.

In both cases, the state of the battery (SoC_{14}) and the action taken (x_{14}) in the final minute are identical. However, the agent receives a vastly different reward. This complicates the learning process, as the agent cannot distinguish between these two histories based solely on the current state.

To resolve this ambiguity, the state space was augmented with two new features:

- A_t^c: The total energy charged within the current quarter up to time t.
- A_t^d: The total energy discharged within the current quarter up to time t.

Justification for Two Separate Features A natural question is why two separate features are used instead of a single feature representing the net energy traded, e.g., $A_t^{net} = A_t^c - A_t^d$. While this would solve the primary ambiguity described above, it would introduce a new one. Consider two distinct scenarios:

- **Scenario A:** The agent charges for 7 minutes and then discharges for 7 minutes. The net energy traded is close to zero ($A_t^{net} \approx 0$), but the total charged and discharged volumes are significant ($A_t^c > 0, A_t^d > 0$).
- **Scenario B:** The agent remains idle for 14 minutes. The net energy traded is zero ($A_t^{net} = 0$), and the charged and discharged volumes are also zero ($A_t^c = 0, A_t^d = 0$).

With a single net feature, these two scenarios would be indistinguishable to the agent. However, Scenario A involves significant energy cycling, which is far more costly from a battery degradation perspective. By keeping A_t^c and A_t^d

separate, the state representation preserves this crucial information about total energy throughput. This allows the agent to correctly correlate its actions to the final reward and provides a foundation for potentially modeling costs like battery degradation in the future.

3.3 Heuristic Policy Definition

While the general model formulation allows for any policy, the baseline heuristic employs a specific, rule-based strategy. This strategy extends the fundamental "buy-low, sell-high" principle by incorporating a trend confirmation mechanism to mitigate risk in volatile markets.

To support this trend analysis, the state observation for the heuristic is augmented with a history of the past 5 prices:

$$S'_t = (SoC_t, p_t, p_{t-1}, p_{t-2}, p_{t-3}, p_{t-4}, p_{t-5})$$

The policy, denoted as $X^{trend}(S'_t|\theta)$, determines the action x_t based on the current price p_t relative to tunable thresholds, and confirms the decision by checking the sign of the previous 5 price observations. The policy is defined as follows:

$$x_t = \begin{cases} r^{chg} & \text{if } p_t \leq \theta^{buy} \text{ and } p_{t-k} < 0 \text{ for all } k \in \{1, \ldots, 5\} \\ r^{dis} & \text{if } p_t \geq \theta^{sell} \text{ and } p_{t-k} > 0 \text{ for all } k \in \{1, \ldots, 5\} \\ 0 & \text{otherwise} \end{cases}$$

where the parameter vector to be optimized is $\theta = (\theta^{buy}, \theta^{sell})$.

This policy ensures that the agent only commits to charging when prices are consistently negative (indicating a surplus) and discharging when prices are consistently positive (indicating a shortage), thereby avoiding rapid fluctuations that could lead to unprofitable trades.

3.4 Evaluation Framework

This section establishes the general evaluation framework utilized throughout this thesis. It defines the standardized methodologies for data partitioning and performance measurement that are necessary to compare different strategies. To demonstrate the practical application of this framework, it is used here to assess the baseline heuristic policy. The discussion begins by detailing the evaluation criteria and concludes with an analysis of the heuristic's performance.

Evaluating the performance of an energy arbitrage agent extends beyond calculating the cumulative financial profit. A policy that yields high total returns but requires excessive battery cycling may be less desirable than a slightly less profitable but more efficient policy due to battery degradation costs. Furthermore, due to the stochastic and seasonal nature of electricity prices, a policy must be robust across different market conditions.

3.4.1 Data Partitioning: Episodic Train-Validation-Test Split

Standard time-series evaluation often utilizes a chronological split (e.g., training on the first 80% of the year and testing on the last 20%). However, in energy markets, this approach is flawed due to strong seasonality. A model trained on summer data might fail to generalize to winter price patterns.

To address this, a Randomized Episodic Split strategy was employed to partition the dataset into three distinct subsets: Training, Validation, and Testing. The full timeline is divided into discrete candidate episodes of N days.

- **Test Set:** First, a fraction (e.g., 20%) of episodes is randomly sampled from the entire year. This set is strictly held out and used only for the final performance reporting.
- **Validation Set:** From the remaining data, a second fraction (e.g., 10%) is randomly sampled. This set is used for hyperparameter tuning and model selection during the development phase, ensuring that configuration decisions do not bias the final test results.
- **Buffer Zones:** To prevent data leakage, a safety buffer is enforced around every selected Test and Validation episode. Data within these buffers is excluded from the Training set to ensure the agent cannot memorize temporal trends immediately adjacent to evaluation periods.
- **Training Set:** All remaining data that is not part of a Test episode, Validation episode, or Buffer zone constitutes the training set.

This method ensures that the evaluation metrics reflect the agent's ability to generalize across diverse market conditions (e.g., high-volatility winter weeks vs. low-volatility summer weekends).

3.4.2 Performance Metrics

To assess the qualitative behavior and risk profile of the agent, the following metrics are computed over the test set:

1. Profitability Metrics

- **Mean Daily Reward (€/day):** The primary Key Performance Indicator (KPI). It represents the total accumulated reward divided by the total number of days in the evaluation period. This normalizes the performance regardless of the simulation length.
- **Mean Reward per Active Quarter (€/q):** The average profit generated only in quarters where the agent actively traded (non-zero reward). This serves as a proxy for efficiency. A low value indicates that the agent is "churning"—performing many trades for marginal gains—which is detrimental to battery life.

2. Reliability and Risk Metrics

- **Success Rate (Active):** The ratio of profitable quarters to total active quarters.

$$\text{Success Rate} = \frac{N_{pos}}{N_{pos} + N_{neg}}$$

where N_{pos} is the count of quarters with positive reward and N_{neg} is the count of quarters with negative reward. A high success rate implies a risk-averse, precise policy.

- **Trap Quarters (Signal Reversal Risk):** This metric quantifies how often the agent was "tricked" by intra-quarter volatility. It counts the specific instances where:

  1. The agent took a position based on a favorable price at the start of the quarter ($t = 0$).
  2. The price reversed direction by the settlement time ($t = 14$), resulting in a financial loss.

  Formally, a "Trap Quarter" occurs if:

$$(A^c_{15} > A^d_{15} \wedge p_0 < 0 \wedge R < 0) \quad \vee \quad (A^d_{15} > A^c_{15} \wedge p_0 > 0 \wedge R < 0)$$

  A high number of trap quarters indicates that the agent fails to account for the uncertainty or volatility of the imbalance price signal.

3. Efficiency and Degradation Metrics The use of the battery is not free. In reality, Lithium-ion batteries degrade with usage; they have a finite cycle life (e.g., 5,000 cycles). A policy that generates high revenue by constantly charging and discharging for minute profit margins ("churning") may actually destroy value by wearing out the battery faster than the revenue generates a return on investment.

To evaluate this trade-off, we introduce metrics that account for the *Levelized Cost of Storage* (LCOS):

- **Daily Cycles:** This measures the physical throughput of the battery normalized by its capacity.

$$N_{cycles} = \frac{\sum_{t=0}^{T} |x_t|}{2 \cdot R^{max} \cdot N_{days}}$$

  where $|x_t|$ is the energy throughput at time t. We divide by $2 \cdot R^{max}$ because a standard "cycle" is defined as one full charge plus one full discharge. A value significantly higher than 1.0 indicates an aggressive strategy that may accelerate degradation.

- **Profit per Cycle:** An important metric that allows us to compare different agents. It is the total profit of the of the agent, divided by the total number of cycles in a certain period.

3.5 Analysis of Heuristic Baseline

3.5.1 Data

The data used is starts on the first of january 2025 and it lasts untill the end of the year. The following results is done on the test set, which is 10 episodes, or 30 days. The battery capacity used is 10MWh. This is a very big battery, but makes things easier as prices are expressed in Euro/MWh.

3.5.2 Parameter Selection

To establish a competitive baseline, the heuristic policy parameters were selected based on a grid search over the training dataset. The goal was to maximize the total accumulated reward while maintaining a reasonable level of risk aversion.

The selected parameters for the trend-based policy $X^{trend}(S_t|\theta)$ are:

- **Buy Threshold (θ^{buy}): 50 €/MWh**. The agent only charges when the price drops below this level, targeting clear negative-price opportunities.
- **Sell Threshold (θ^{sell}): 150 €/MWh**. The agent discharges only when prices exceed this level, ensuring a minimum spread.

3.5.3 Performance Evaluation

The heuristic policy was evaluated on the held-out test set using the standardized metrics defined in the evaluation framework. The quantitative results are summarized in Table 1. Here an active quarter is a quarter where some (dis)charging from the net is done.

Metric	**Value**
Profitability	
Mean Daily Reward	1110.23 €/day
Mean Reward per Active Quarter	62.93 €/q
Profit per Battery Cycle	1537.43 €/cycle
Reliability & Activity	
Profitable Quarters (N_{pos})	410
Unprofitable Quarters (N_{neg})	111
Success Rate (Active)	78.69%
Total Battery Cycles (N_{cycles})	41.12

Table 1: Performance Metrics for the Heuristic Baseline Policy

3.5.4 Discussion of Results

The heuristic achieves a commendable **Mean Daily Reward** of 1110.23 €, demonstrating that a simple rule-based approach can effectively capture the primary arbitrage opportunities in the imbalance market. The policy exhibits a strong **Success Rate** of 78.69%, indicating profit can quite easily be made on this market. So there is a lot of potential for more advanced heuristics/agents.

Conservatism and Absolute Thresholds As indicated by the Profit per Battery Cycle (1537.43 €), the heuristic is highly selective. While this minimizes battery wear, it also reveals a significant limitation: the policy is overly conservative. This behavior is clearly visible in Figure 1.

Between the 9th and 11th of January, the imbalance price remains consistently positive, fluctuating between approximately 50 €/MWh and 300 €/MWh. During this 48-hour window, the agent's State of Charge (SoC) remains at zero, and almost no trading activity occurs. Because the heuristic is bound by a fixed absolute buy threshold (θ^{buy}), it refuses to "invest" in energy at positive prices, even when clear arbitrage spreads exist.

Lack of Leverage and Arbitrage Potential This highlights the agent's inability to utilize *leverage*. In this context, leverage refers to the strategic decision to buy energy at a non-optimal (positive) price with the intent of selling it at a significantly higher price later. The heuristic lacks the foresight to recognize that a price of 80 €/MWh can be "cheap" if a spike to 400 €/MWh is imminent.

By only acting on absolute price extremes, the heuristic effectively ignores a large portion of the market's volatility. This results in the battery sitting idle during periods of high price levels where substantial revenue could be generated through relative price arbitrage. These observations justify the move toward Reinforcement Learning agents, which have the potential to learn a more dynamic policy that adapts to relative price trends rather than relying on rigid, absolute thresholds.

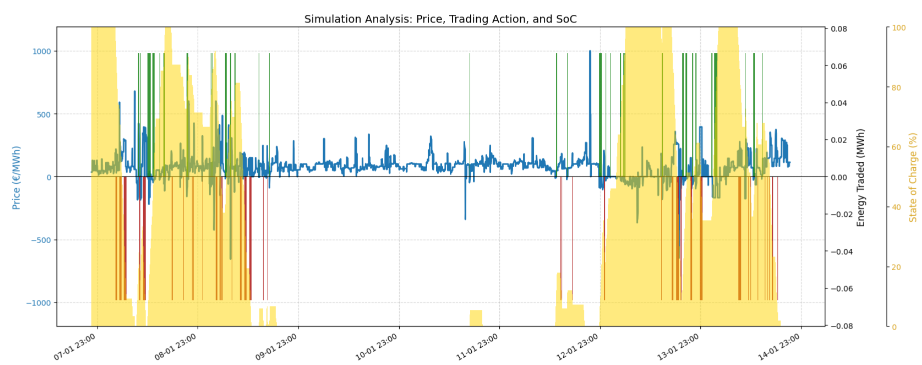

Figure 1: Minute-by-minute simulation trace of the heuristic policy.